A Temporal Heterogeneous Biomedical Knowledge Graph for Quasi-Prospective Clinical Trial Advancement Prediction

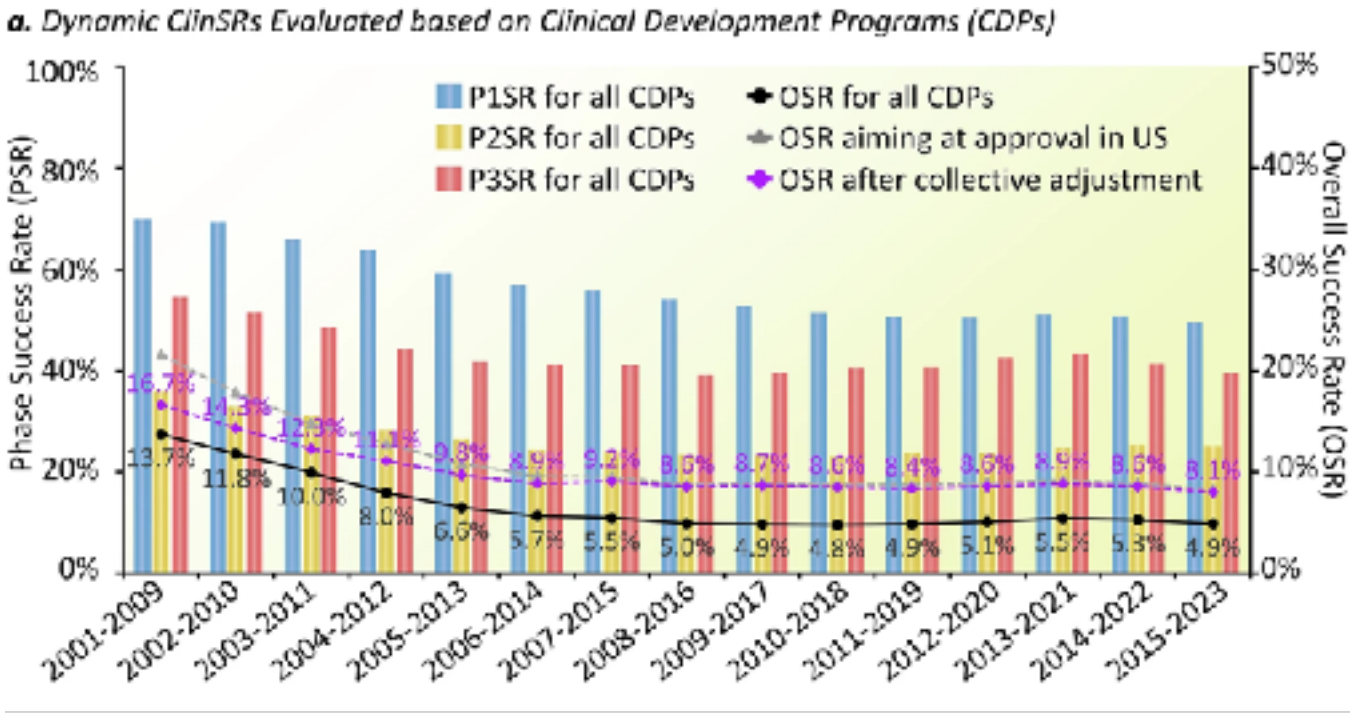
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Phase II→III success stays low: links to inadequate target–disease evidence

Part 1 of 2 · the trend



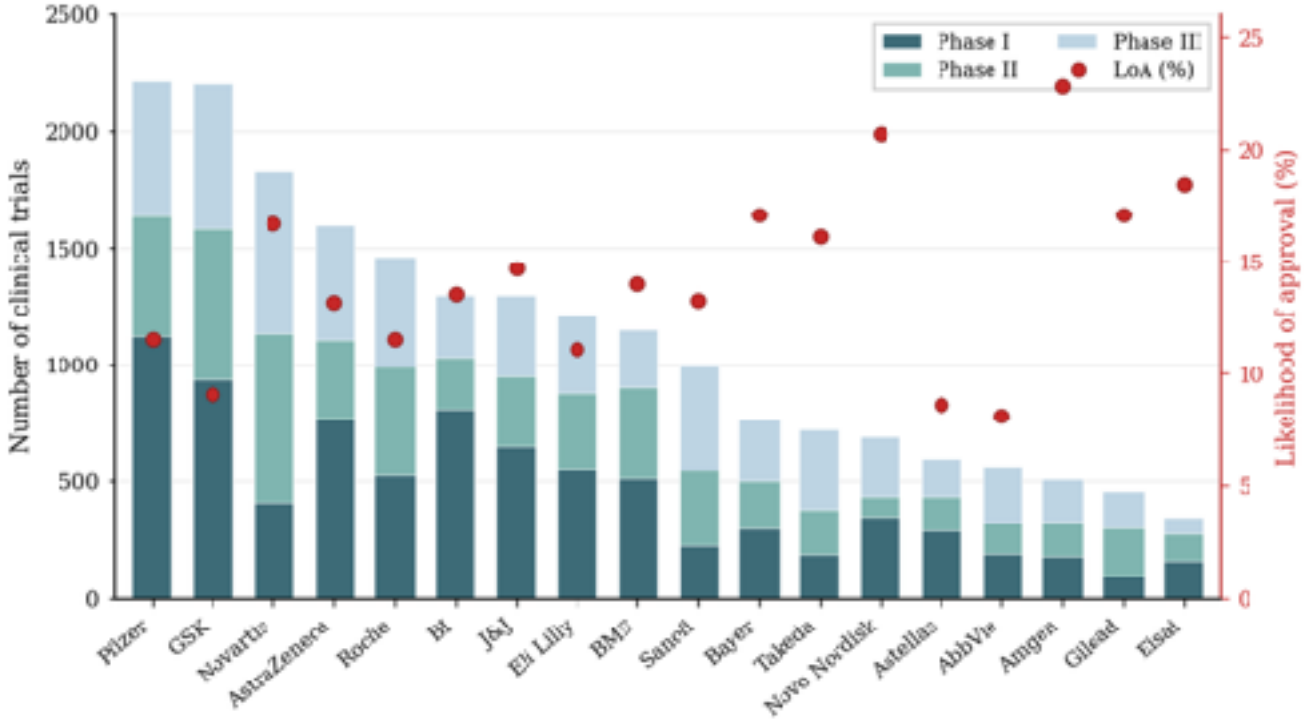
Phase success rates over 2001–2023 rolling windows (CDPs).

- *Eroom's Law* persists
  - phase success rates keep declining despite decades of progress.
- the cost of developing a new, FDA-approved pharmaceutical drug doubles roughly every nine years
  - Phase II→III success rate remain low
    - 40–50% of Phase II efficacy failures trace back to inadequate target–disease linkage.

Zhou et al. 2025, Nat. Commun. 16:9537 (Fig. 2a). Scannell et al. 2012, Nat. Rev. Drug Discov. 11:191.

With 100+ parallel trials per sponsor, re-ranking Phase II assets is a practical lever

Part 2 of 2 · the portfolio

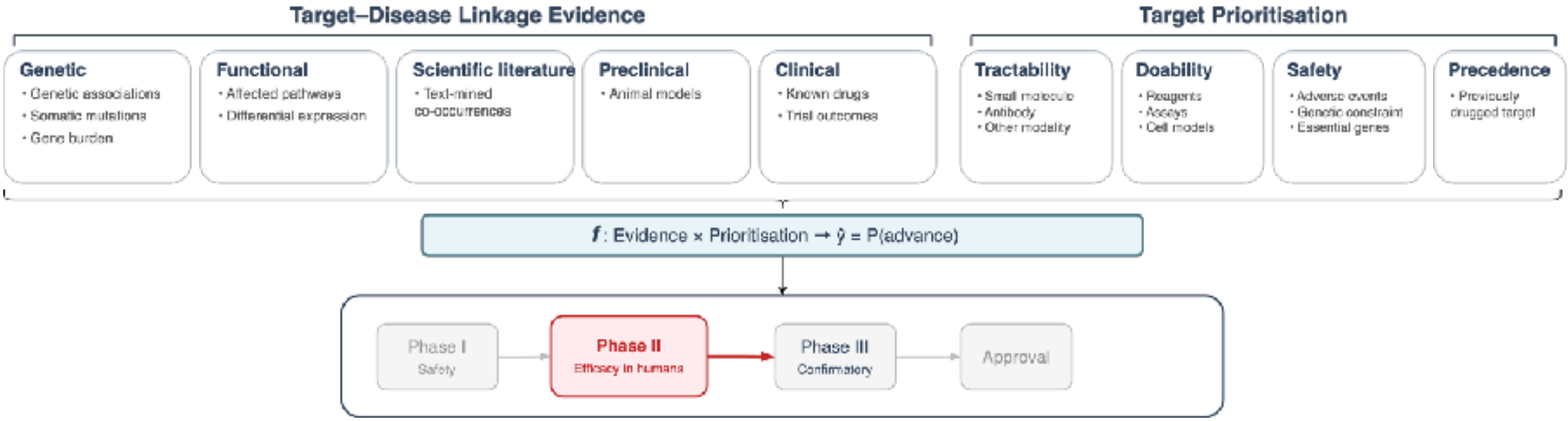


Clinical-trial volume and likelihood of approval by sponsor (top-20 pharma).

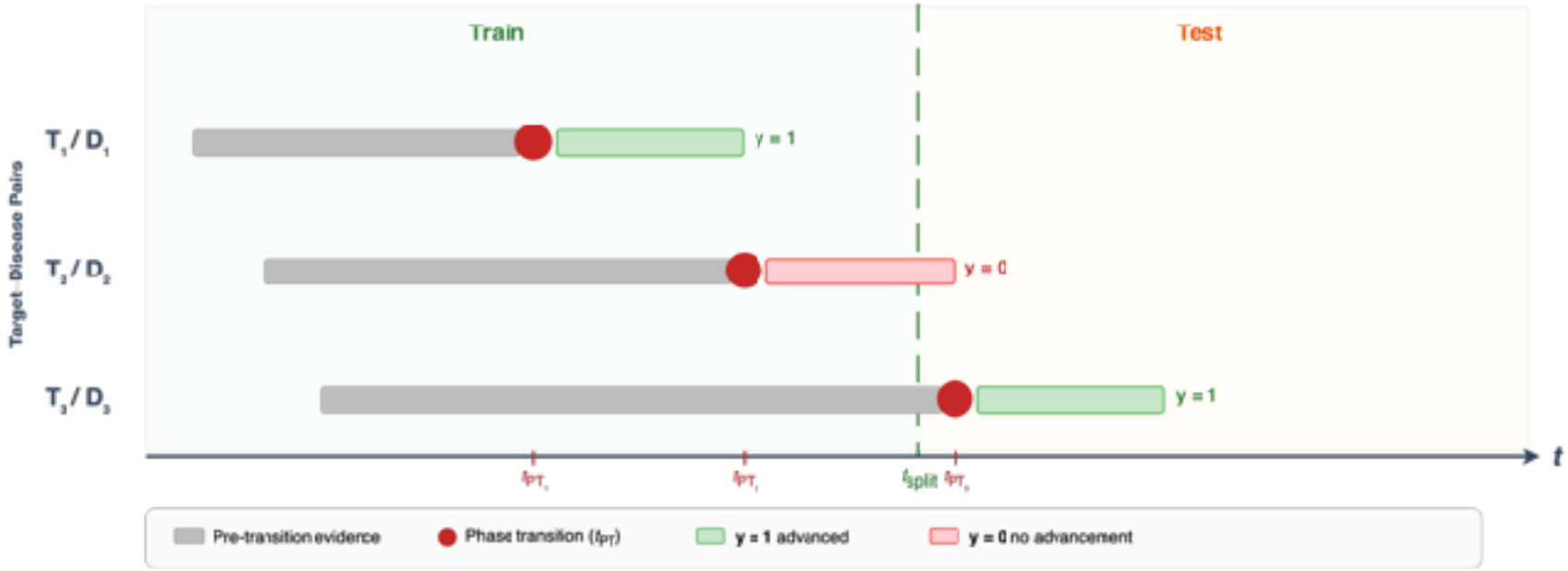
- Top-20 pharma each run 100+ active trials in parallel.
  - Internal and external competition across pharma competitive landscape
- Internal Phase II→III re-ranking of clinical trials for resource allocation and success prioritisation

Schuhmacher et al. 2025, Drug Discov. Today 30:104291. Wouters et al. 2020, JAMA 323:844. DiMasi et al. 2016, J. Health Econ. 47:20.

Problem definition: rank Phase II target–disease pairs by likelihood of advancement, integrating diverse evidence and target-prioritisation signals.



Quasi-prospective evaluation: Emulate the pharma go/no-go decision



Restrictive feature view

- TD pairs entering phase 2 trials for the first time
  - Reduce feature inflation risk
    - New evidence derived from successful / failure of trial

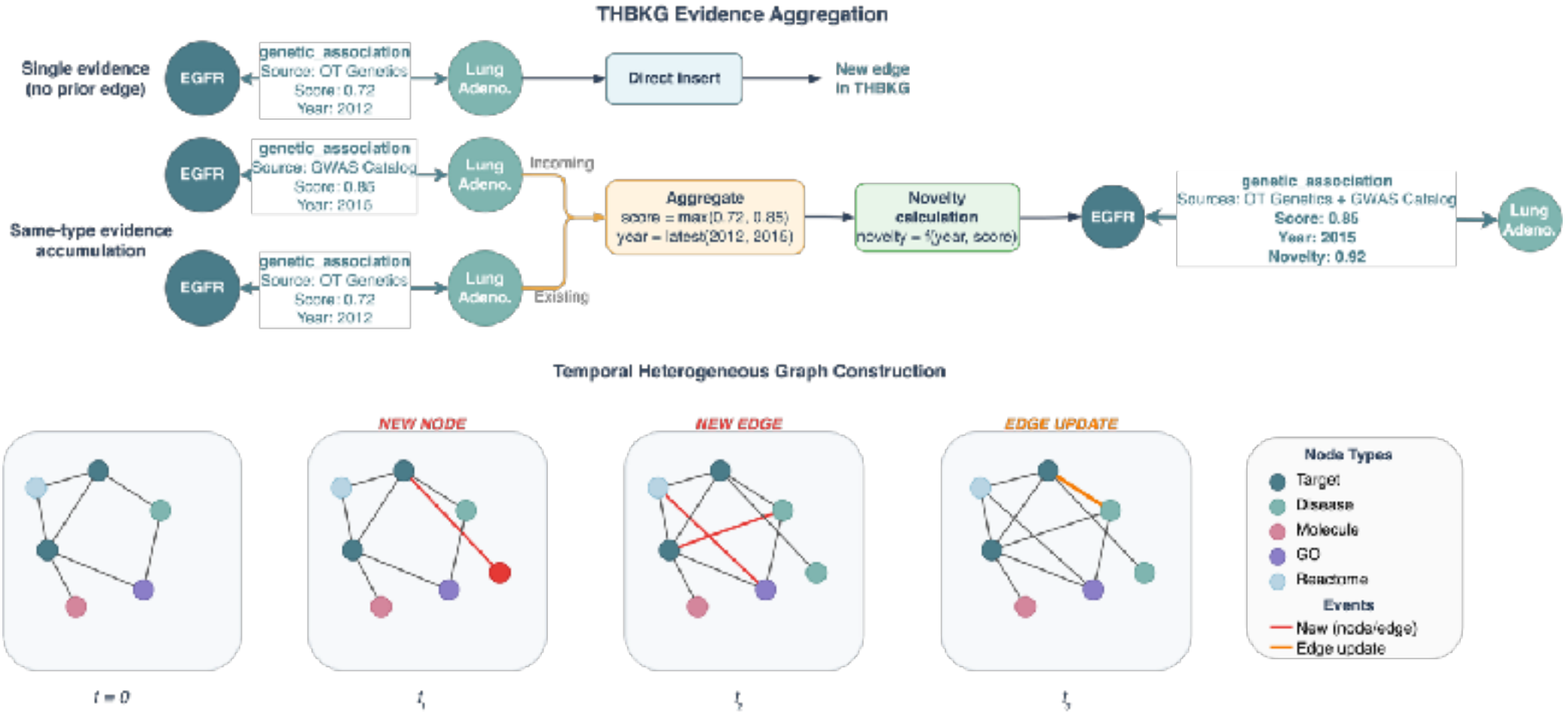
Retrospective validation

- Time machine approach
  - Train model on historical data to assess performance in future

Decision-split mechanism per Czech et al. 2024. Quasi-prospective paradigm also adopted by Aliper et al. 2023 (InClinico).

The graph grows over time: evidence is merged into edge strengths as it arrives.

From timestamped evidence to temporally-resolved strength estimates.



Top: *per-source evidence aggregation (single insert → same-type merge).* ***Bottom:*** *graph events across years (new node, new edge, edge update).*

Temporal by design: biomedical KG that reconstruct target–disease linkage at any point in time

Nodes by type

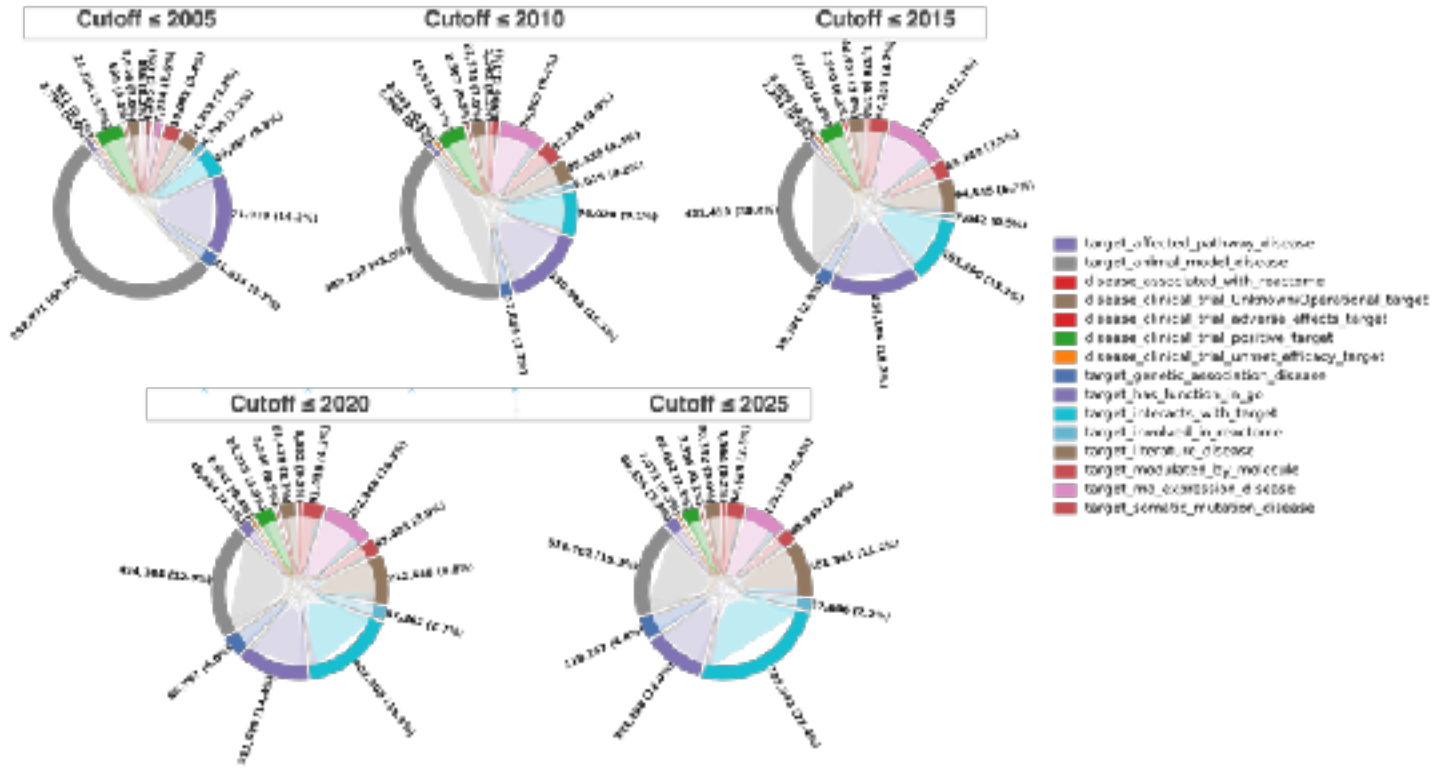
Type	Count
Target	19,316
Disease	11,403
GO term	14,686
Molecule	2,703
Pathway	1,218
Total	**49,326**

Edges by relation type

Relation	Edges	Relation	Edges
PPI (interacts_with)	**659,444**	trial: unknown / operational	52,738
animal_model	442,082	drug modulation (modulated_by)	45,416
GO function (has_function_in)	256,090	affected_pathway	38,804
literature	237,795	target → pathway (involved_in)	36,972
rna_expression	171,555	Trial:ongoing	134,410
genetic_association	82,408	disease → pathway (assoc_with)	3,639
somatic_mutation	63,557	*+ 2 trial outcome relations*	*6,355*
trial: positive	56,190		

Total temporal edges ≈ 2,187,455 *(+ 3 static edge types)*

Edge population at five cumulative temporal cutoffs

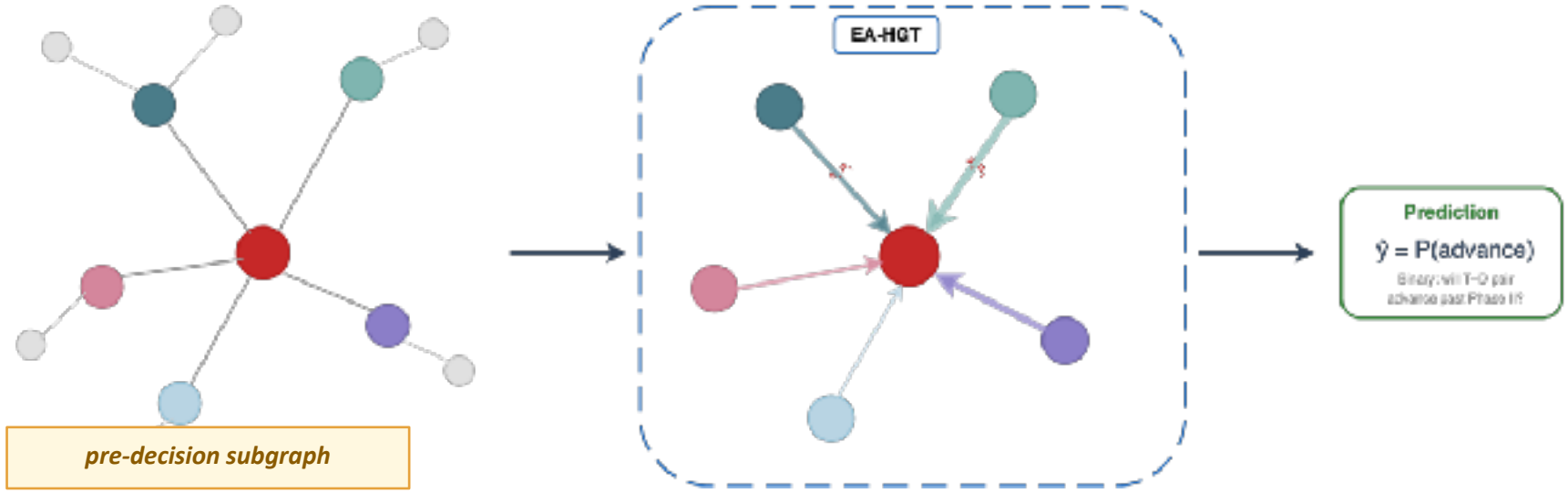


Open Targets release 26.03. Narganes-Carlón et al. 2024.

EA-HGT scores each pair from its pre-decision subgraph only: attending to edge type and strength

Temporal alignment is handled upstream by per-instance masking

$$Attention(s, e, t) = \text{softmax}\left(Q_{\tau(t)} W^{ATT}_{\phi(e)} K_{\tau(s)}^{T} / \sqrt{d_k} \odot g(\text{score, novelty}) \right) W^{MSG}_{\phi(e)} V_{\tau(s)}$$



Attends to: node features, node type, edge type, edge strength.

Ranking task: pairs scored by $\hat{y}$ = *P(advance)*, then ranked.

RS@N

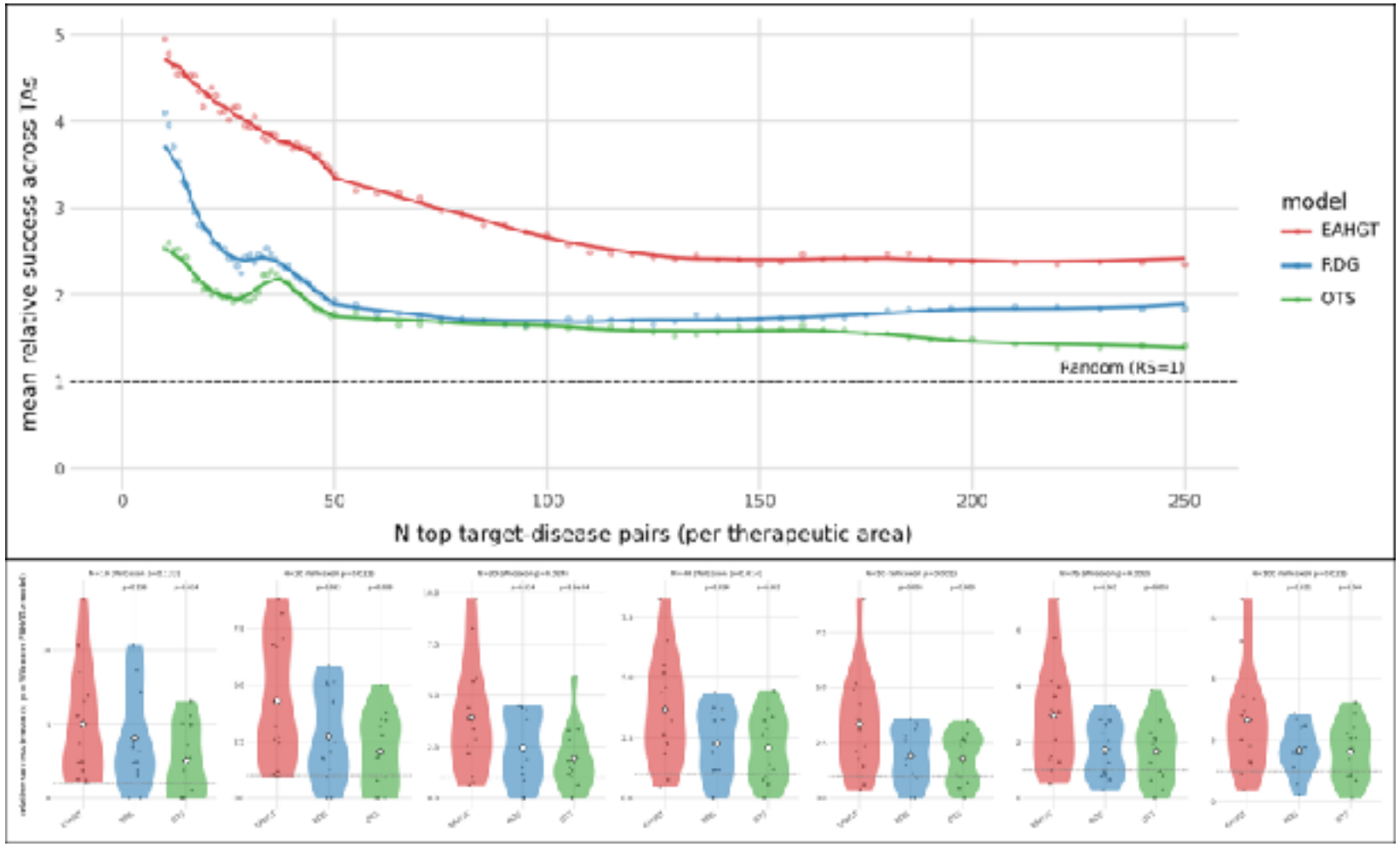
$$P(\text{advance} \mid \text{rank} \le N) \;/\; P(\text{advance} \mid \text{rank} > N)$$

enrichment of advancers in the top-N pairs vs the rest

EA-HGT exceed SOTA baseline in RS@N across the actionable range.

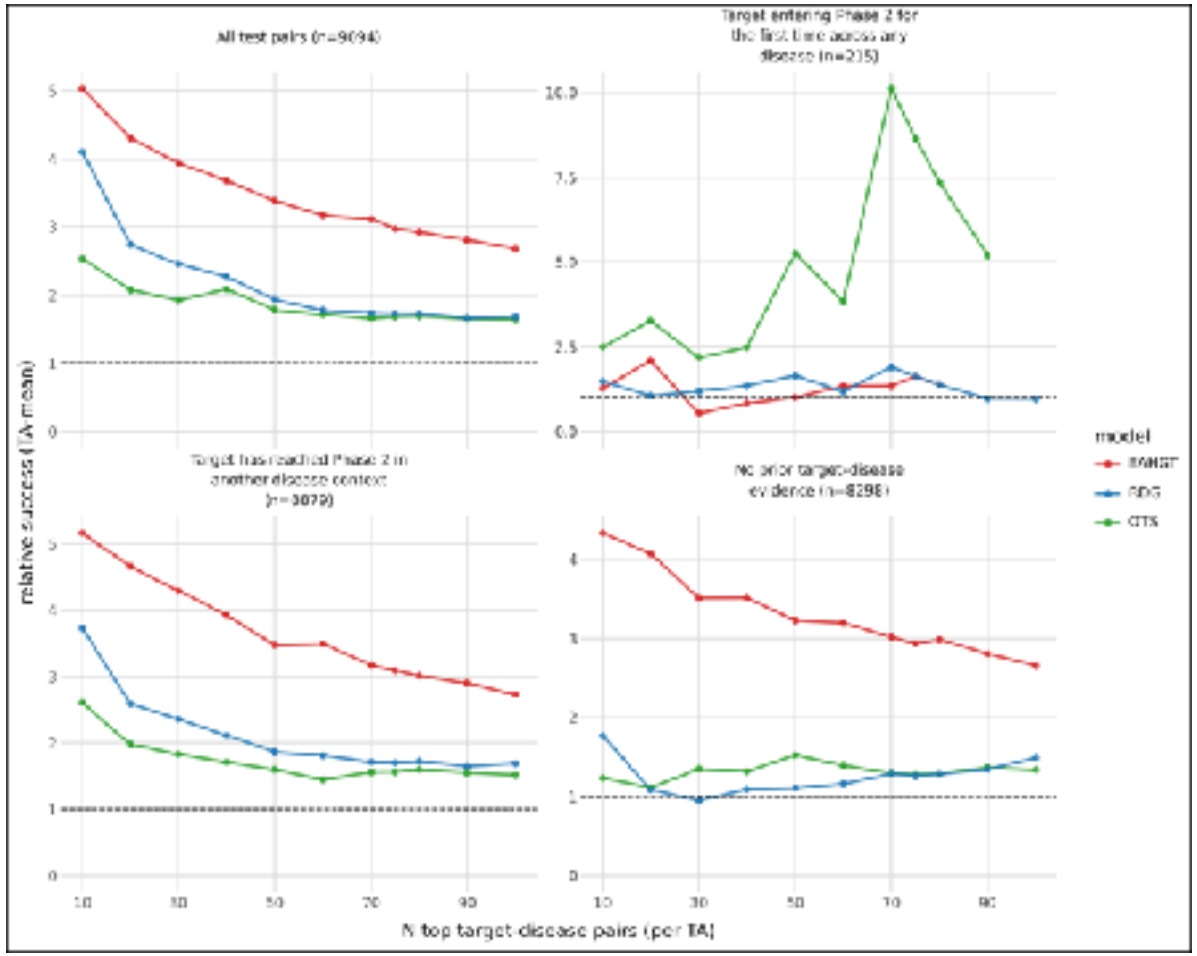
Comparators

Model	Description
OTS	Open Targets composite association score Reconstructed
RDG	Ridge regression over longitudinal Open Targets features (Czech et al.);
EA-HGT	Edge-aware HGT with score and novelty; proposed full model.

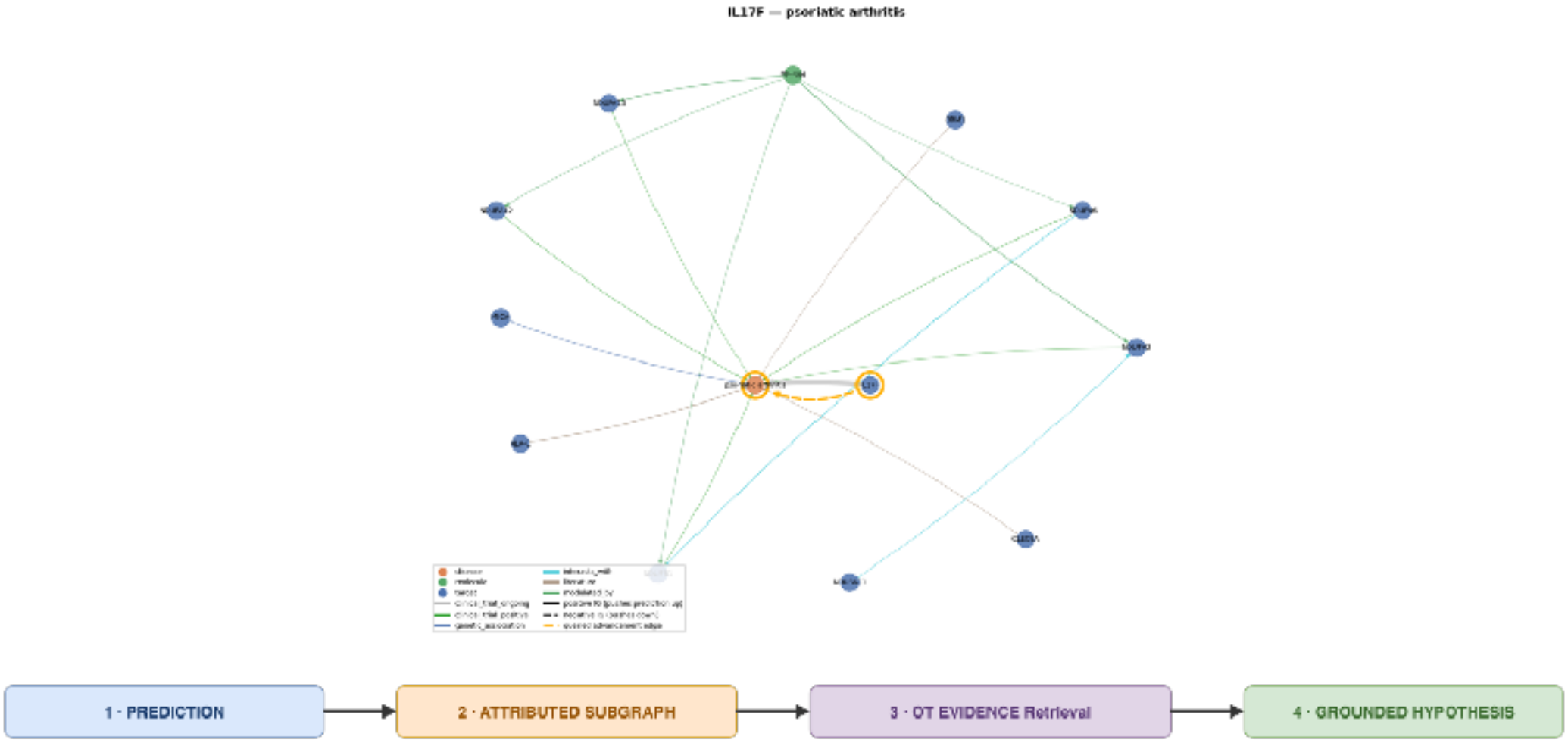


TA-averaged across 13 therapeutic areas. Open Targets 26.03.

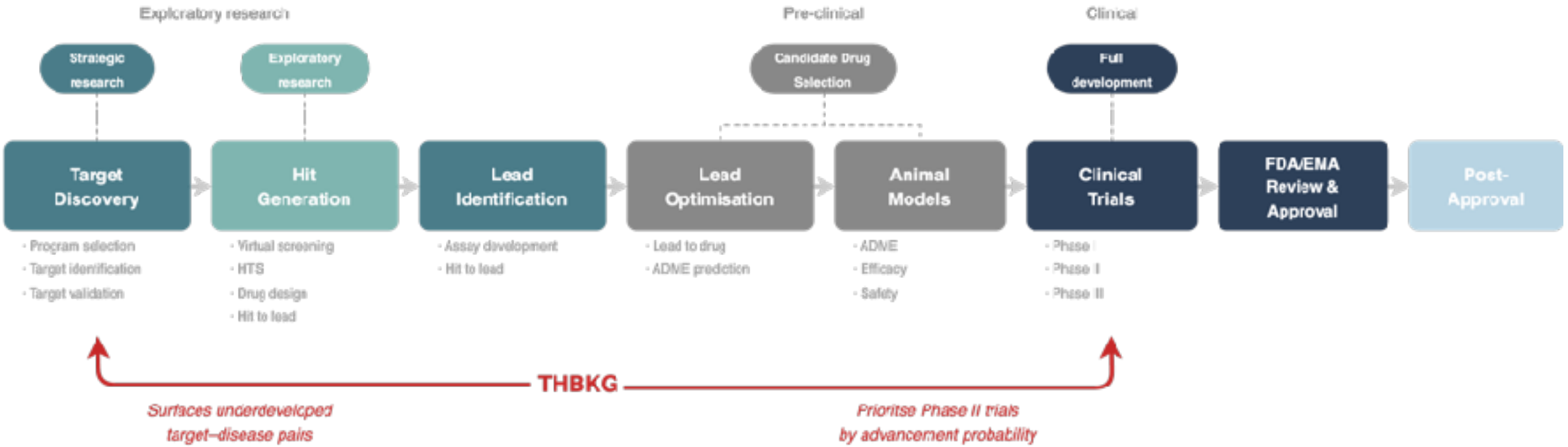
The advantage concentrates where there is no direct evidence and where prior target clinical evidence exists



Integrated gradient attribution recovers the multi-hop subgraph structure that supports advancement prediction



THBKG + EA-HGT enables two portfolio decisions: derisking existing Phase II assets and discovering underdeveloped TD pairs.



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Zhou, Y., Zhang, Y., Xu, H., et al. (2025). Dynamic clinical trial success rates for drugs in the 21st century. Nature Communications, 16(1), 9537.

Appendix

Appendix A1: Full ablation table — 7 conditions plus 2 references.

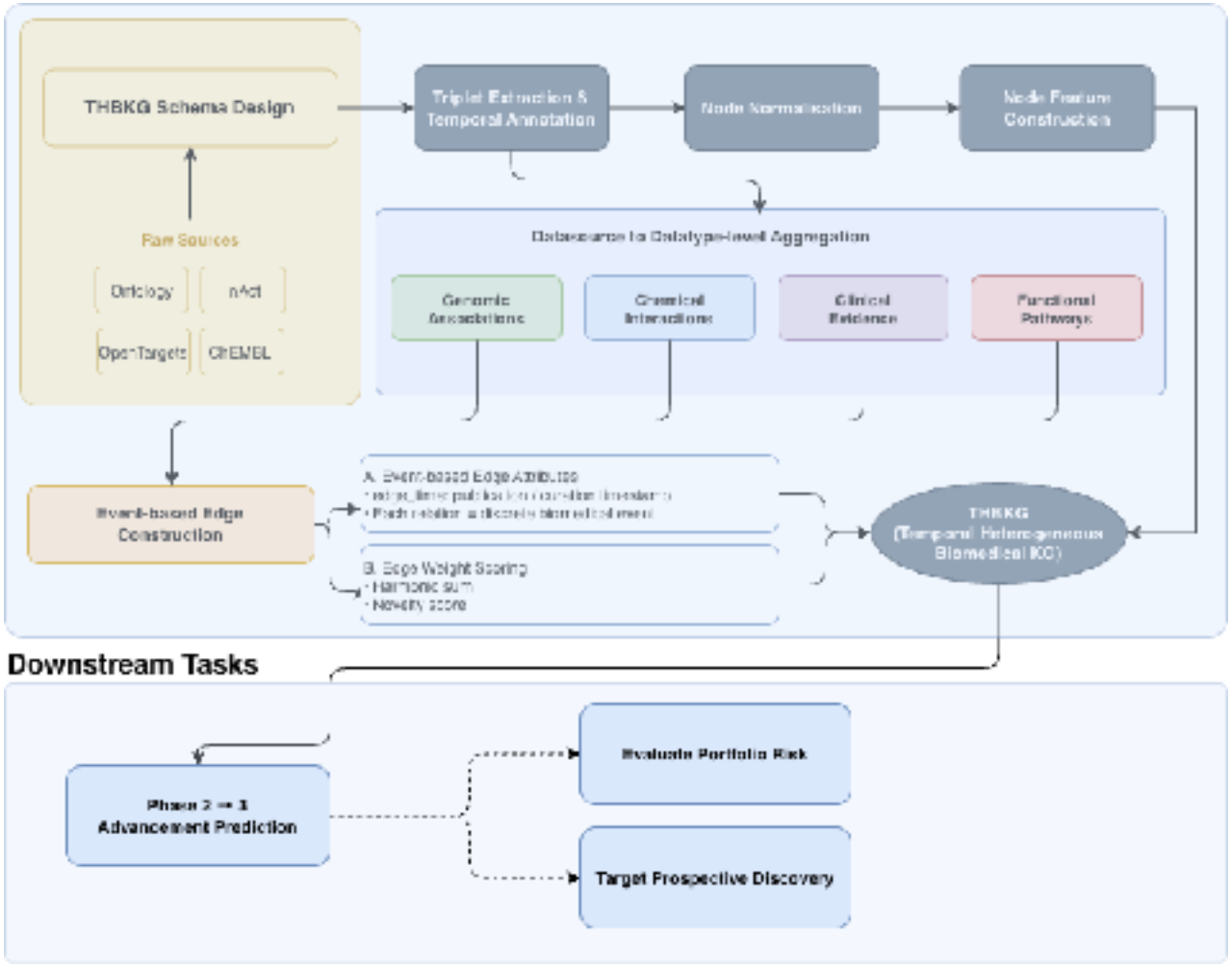
Model	RS@10	RS@30	RS@50	RS@100	AUROC	AP
OTS	2.53	1.92	1.78	1.64	0.49	0.09
RDG	4.10	2.45	1.94	1.68	0.64	0.12
HGT	0.98	1.25	0.98	0.99	0.55	0.09
GATv2	2.18	1.97	2.09	2.29	0.59	0.12
RGCN	3.26	3.59	3.52	3.20	0.52	0.12
CompGCN	1.09	0.89	0.82	1.28	0.62	0.13
EA-HGT (s)	4.35	3.51	3.84	3.83	0.64	0.18
EA-HGT (n)	0.43	1.45	1.58	1.34	0.61	0.10
EA-HGT (s, n)	**6.02**	**5.84**	**6.19**	**4.82**	**0.59**	**0.19**

What this isolates

- Plain HGT is the weakest model in the table.
- Add cumulative score: RS@100 jumps 0.99 → 3.83.
- Add novelty too: 3.83 → 4.82.
- Heterogeneous attention is necessary but not sufficient — the lift comes from coupling it to instance-level evidential weight.

TA-averaged RS; AUROC and AP over all evaluation pairs. Open Targets 23.06.

Appendix A2: Multi-hop graph propagation survives temporal masking where direct-edge features cannot.

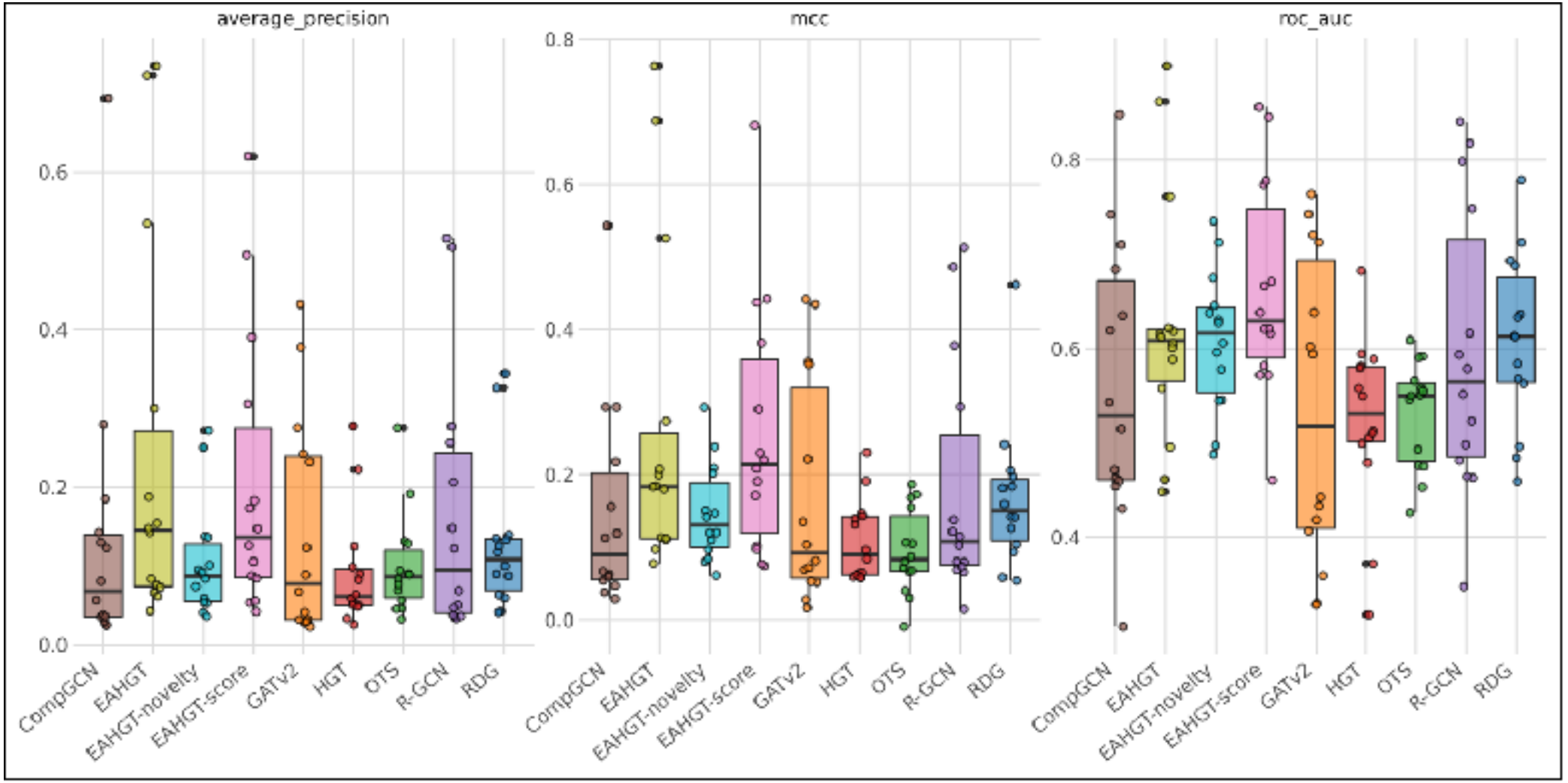


THBKG construction pipeline.

The structural backbone

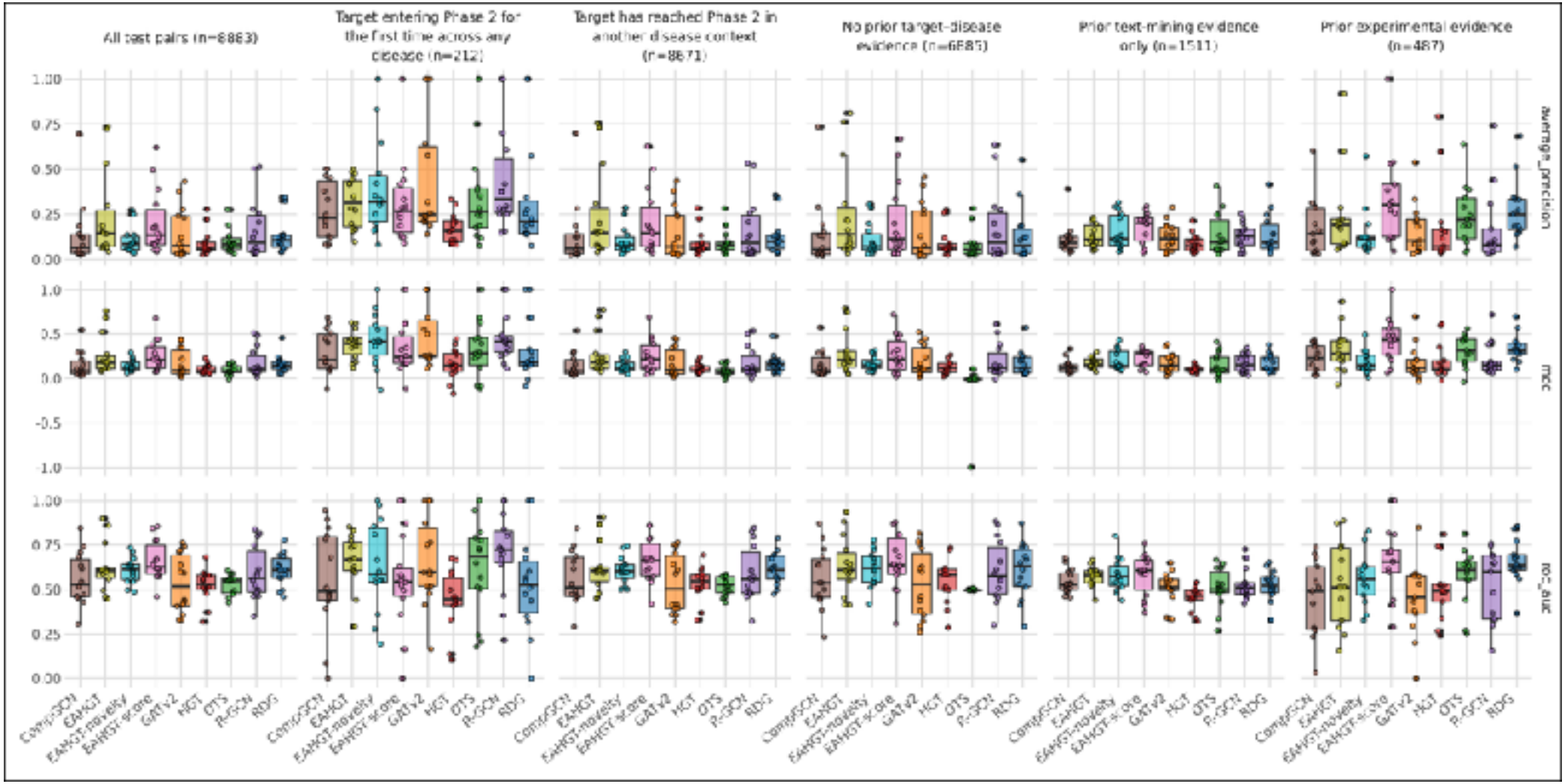
- Direct target–disease evidence is what gets masked.
- But the graph's structural edges — protein–protein interactions, pathway membership, GO annotations, drug modulation — mostly persist.
- The encoder still receives non-trivial input even when no direct evidence remains.
- Multi-hop neighbourhoods become the predictive substrate.

Appendix A3: Per-therapeutic-area classification metrics — EA-HGT has the highest median AP.



Box = per-TA distribution; points = individual TAs. Decision-aligned, Open Targets 23.06.

Appendix A4: Stratum × therapeutic-area classification metrics.



AP and AUROC, decision-aligned, Open Targets 23.06.

Appendix A5: EA-HGT (s, n) hyperparameter configuration.

Encoder	
Hidden / output dim	128
Attention heads	2
HGT layers	2
Dropout (encoder)	0.2
Edge feature dim	2 (s_{ij}, n_{ij})
Neighbour sample (L1 / L2)	20 / 10
Decoder	
MLP layers	256 → 128 → 64 → 1
Dropout (decoder)	0.1
Training	
Loss	LambdaLoss (k=50, σ=1.43, ungrouped)
Optimiser	AdamW
Learning rate	4.35×10^{-4}
Weight decay	1.83×10^{-3}
LR scheduler	CosineAnnealingLR ($\eta_{min} = 10^{-6}$)
Epochs / batch	50 / 512
Gradient clip / early stop	max_norm 1.0 / patience 5 (NDCG@50)

Single NVIDIA A100, QMUL Apocrita HPC.